



Unsteady Magnetohydrodynamic Sutterby Penta-Hybrid Nanofluid Flow Over an Extending Catalytic Surface for Wastewater Treatment

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Abstract: An unsteady magnetohydrodynamic boundary-layer model was developed for a Sutterby penta-hybrid nanofluid flowing over an extending catalytic surface, with particular emphasis on transport mechanisms relevant to wastewater treatment. Water was employed as the base fluid, while polystyrene, poly (acrylic acid), poly (acrylic acid)-block-polystyrene (PAA-b-PS), cerium oxide, and copper oxide were incorporated to represent complementary functionalities associated with colloidal stabilization, contaminant adsorption, photocatalytic degradation, and antimicrobial activity. The coupled conservation equations governing momentum, energy, and species concentration were formulated in Cartesian coordinates, together with a Poisson equation for the pressure field. Appropriate similarity transformations were subsequently introduced to reduce the governing partial differential equations to a coupled system of nonlinear ordinary differential equations. Surface-catalyzed contaminant degradation was represented through reaction boundary conditions parameterized by Damköhler numbers. The resulting boundary-value problem was solved numerically using the MATLAB bvp5c solver. The computed results demonstrated that the hydrodynamic, thermal, and concentration boundary layers were strongly governed by the combined effects of magnetic forcing, Sutterby rheological behavior, nanoparticle loading, unsteadiness, and reaction kinetics. The effects of the magnetic field, Sutterby rheological parameters, nanoparticle loading, reaction kinetics, and pressure variations on the velocity, temperature, and contaminant-concentration distributions were systematically evaluated. The numerical results demonstrated that the coupled effects of magnetohydrodynamic forcing, Sutterby rheology, and penta-hybrid composition substantially modified momentum, thermal, and mass transport within the boundary layer. In particular, appropriate combinations of the governing parameters were shown to intensify thermal and solutal transport and promote contaminant degradation at the catalytic surface. Pressure variations were additionally demonstrated to influence boundary-layer development and species transport, thereby affecting the predicted contaminant-removal characteristics. These findings establish a theoretical framework for understanding coupled magnetohydrodynamic, non-Newtonian, heat-transfer, and reactive mass-transfer phenomena in multifunctional nanofluid systems and provide potential guidance for the development and optimization of advanced wastewater-treatment processes.

Keywords: Sutterby fluid; Penta-hybrid nanofluid; Magnetohydrodynamics; Wastewater treatment; Catalytic degradation; Heat transfer; Boundary layer flow; MATLAB bvp5c

1 Introduction

Water pollution has become one of the most significant ecological issues globally as a result of swift industrial growth, urban expansion, and extensive farming practices. The persistent release of organic dyes, heavy metals, pharmaceutical contaminants, and pathogenic microorganisms into aquatic environments has generated an immediate demand for effective and sustainable treatment solutions. Despite the prevalent use of traditional wastewater treatment techniques, its inadequate efficacy in eliminating stubborn and intricate pollutants has spurred the advancement of sophisticated treatment systems. In this framework, treatment systems utilizing nanotechnology have attracted significant interest due to their extensive surface area, superior adsorption capabilities, catalytic efficiency, and enhanced thermal and mass transfer properties. Zhao and Tu [1] demonstrated electrochemical–membrane integration for hydrazine wastewater treatment that are thermally regulated and the uncontrolled industrial discharge of highly toxic chemical pollutants such as hydrazine into water bodies poses critical ecological risks, underscoring the urgent need for effective and sustainable strategies to protect water resources from the escalating burden of anthropogenic contamination. The single-parameter treatment strategies are poorly equipped to handle the accelerated pace of industrial discharge into surface and groundwater bodies, which has created complex, multi-pollutant scenarios. Beyond conventional chemical oxygen demand and suspended solids, the multifunctional remediation frameworks are reinforced due to the emerging contaminants such as endocrine-disrupting compounds, antibiotic residues, and synthetic dyes now posing compound threats to aquatic biodiversity and human health.

Saxena [2] explored the consequences of industrialization and urbanization on the environment. Environmental systems lose their natural balance due to modern practices such as industrialization on a large scale, urban development and agricultural expansion which all lead to a continuous discharge of harmful substances into freshwater bodies. Yunus et al. [3] explored environmental remediation approaches through nanotechnology. The harmful substances cause severe danger to human health and to the environment. The harmful substances are obtained from modern practices which include organic dyes, heavy metals, pharmaceutical products, microbial pathogens and other biological waste. For treating pollutants, the classical methods commonly include sedimentation, filtration, absorption, coagulation and flocculation, bioremediation, activated sludge, pyrolysis, etc. The standard methods for handling persistent pollutants and these methods fail to purify complex tasks because they require low-reactivity materials. Kato and Kansha [4] highlighted the challenges associated with conventional treatment processes and reviewed treatment for industrial wastewater. Currently, there is a shortage of better wastewater treatment for high levels of pollutants, enabling scalability and sustainability throughout the extended period.

This inadequacy is solved by nanotechnology. Shamshad and Ur Rehman [5] explored the technologies that presented innovative and sustainable approaches for wastewater treatment. Nanotechnology emerges as an innovative approach offering high surface area, tunable surface chemistry, and strong catalytic activity for treating wastewater, as it is a promising, efficient, and cost-effective methodology in wastewater treatment. Aksoy [6] studied sustainable heat transfer enhancement of nanofluids, and Nour et al. [7] reviewed nanomaterials and nanofluids for environmental applications and highlighted their multifunctional roles. Nanofluids are engineered colloidal suspensions of nanoscale particles dispersed in carrier fluids. This carrier fluid exhibits the ability to transport thermal energy and to tune the physicochemical properties. Their high surface area and interfacial characteristics have expanded their applications beyond heat transfer systems toward environmental remediation. Ranjbarzadeh and Sappa [8] conducted numerical and experimental approaches on fluid flow and heat transfer in porous media, while Rahman et al. [9] reviewed nanofluid preparation and characteristics. Their superior thermal conductivity and diffusivity make nanofluids beneficial not only for thermal engineering but also for reactive flow-based purification configurations in porous media. Gnanasekaran et al. [10] synthesized and characterized metal oxide nanoparticles for degradation of textile dyes, while Gupta et al. [11] developed a cerium oxide–copper oxide heterostructure, indicating the potential of combined cerium and copper oxide systems for efficient wastewater remediation and demonstrating enhanced cationic dye degradation and microbial inactivation. Metal oxide nanomaterials function as effective adsorbents for heavy metals and act as photocatalysts to degrade organic pollutants, dyes and pathogens, providing a sustainable alternative for removing persistent contaminants. Among metal oxide nanomaterials, copper oxide and cerium oxide are impactful agents for remediation of wastewater. Copper oxide nanoparticles are productive against both organic pollutants and pathogenic microorganisms due to their photocatalytic activity and broad-spectrum antimicrobial performance.

Xu et al. [12] reviewed advances in cerium dioxide nanomaterials, their properties, and applications. Cerium oxide nanoparticles exhibit a unique oxygen storage capacity that facilitates reactive oxygen species scavenging and efficient contaminant degradation by virtue of $\text{Ce}^{3+}/\text{Ce}^{4+}$ redox transition. When these two metal oxides are combined, their catalytic and antibacterial activities interact synergistically, yielding degradation kinetics and disinfection efficiency that neither material achieves in isolation. Dey et al. [13] and Shaji et al. [14] reviewed polymeric nanocomposites and highlighted their role in pollutant adsorption, degradation, and contaminant removal for wastewater treatment and environmental remediation. Wilson et al. [15] demonstrated the application of PAA-b-PS as a polyelectrolyte block co-polymer stabilizer for systems based on nanoparticles. The importance of managing

the surface charge of nanoparticles and their colloidal properties.

The proposed penta-hybrid nanofluid consists of five elements with complimentary activities in wastewater treatment. Polystyrene acts as structural and hydrophobic adsorption support and poly (acrylic acid) offers functional groups to facilitate pollutant contact and stabilization of the nanoparticles. PAA-b-PS combines the complementing features of both polymers and demonstrates an increased dispersion stability. Cerium oxide possesses redox activity via the $\text{Ce}^{3+}/\text{Ce}^{4+}$ transition and enables degradation of contaminants. Copper oxide possesses photocatalytic and antibacterial properties. The selected penta-hybrid formulation is thus intended to incorporate stabilization of nanoparticles, adsorption of contaminants, catalytic degradation and antibacterial activity in a single wastewater treatment medium.

Tawalbeh et al. [16] reviewed non-Newtonian nanofluids, while Ullah et al. [17] studied Sutterby nanofluid flow characteristics, providing insights into the rheological behavior of complex nanofluids. From the perspective of continuum fluid mechanics, the multi-component nanofluids, particularly those containing polymer chains and mixed nanoparticle assemblies, regularly exhibit non-Newtonian flow behavior characterized by shear-rate-dependent viscosity, viscoelastic memory, and stress relaxation. Shende et al. [18] investigated the impact dynamics of nanoparticle dispersions in viscoelastic non-Newtonian fluids. Saqib et al. [19] investigated energy transport and entropy analysis for non-Newtonian Sutterby nanofluid flow through intersecting planes. For these complex rheological features, standard Newtonian models are inherently incapable of representation. Among the available non-Newtonian constitutive frameworks, the model should describe both shear-thinning and shear-thickening responses across a wide range of strain rates; the Sutterby fluid model is especially well-suited and has been validated extensively for polymer solutions and hybrid nanofluid flows in engineering applications. Almetwally et al. [20] studied pollutant dispersion and nanoparticle dynamics in magnetized bioconvection for sustainable water treatment. The nanofluid transport is also achieved by external magnetic fields which is an additional degree of physical complexity for transport processes. Zaidi et al. [21] explored magnetic field applications and their potential in wastewater treatment. When a transverse magnetic field is imposed on an electrically conducting nanofluid, it results in a Lorentz force that acts as a resistive body for opposing the primary flow direction, thereby modifying the development of velocity, thermal, and concentration boundary layers. Ma et al. [22] discussed catalytic membrane-based oxidation–filtration systems for water remediation, highlighting the importance of mass transfer, surface reactions, and catalytic efficiency, while Xu et al. [23] reviewed advances in catalytic materials for wastewater treatment, focusing on design strategies and reaction mechanisms. In the context of catalytic wastewater treatment, magnetohydrodynamically induced flow retardation can be strategically exploited to increase the residence time of contaminant species within the active reaction zone near the nanoparticle-coated surface, thereby improving the overall catalytic conversion efficiency. Catalytic purification systems break down the pollutant at the nanoparticle surface through the kinetics of chemical reactions. The purification performance is determined by the competition between convective transport from one surface to another, molecular diffusion, and heterogeneous surface reaction rates. Gundagani and Babu [24] investigated the effects of magnetic fields on nanofluid flow over a stretching sheet embedded in a porous medium, considering variable viscosity and convective boundary conditions.

This study addresses this gap by developing a comprehensive Sutterby penta-hybrid nanofluid model incorporating magnetohydrodynamic effects, Poisson pressure distribution, viscous dissipation, heat generation, thermal and solutal relaxation, and catalytic surface reactions. The effects of key dimensionless parameters on velocity, temperature, concentration, and pressure distributions are systematically investigated, providing useful insights into transport mechanisms relevant to catalytic wastewater treatment systems. The mathematical framework consists of coupled nonlinear partial differential equations governing momentum, heat, mass, and pressure transport. Similarity transformations reduce these equations to nonlinear ordinary differential equations, which are solved numerically using MATLAB's bvp5c solver. Despite extensive research on hybrid nanofluids, studies integrating polymeric stabilizers and metal oxide nanoparticles for wastewater treatment remain limited.

The originality of this study resides in the following:

- Creating a Sutterby penta-hybrid nanofluid model containing poly (acrylic acid)/polystyrene/PAA-b-PS/cerium oxide/copper oxide nanoparticles designed for wastewater treatment.
- Incorporating a Poisson pressure distribution into similarity-transformed nanofluid flow models, a method seldom explored in purification research.
- Integrating chemical reactions to accurately represent pollutant-degrading pathways.
- This approach offers a novel computational platform for optimizing nanofluid-assisted catalytic wastewater treatment systems, facilitating improved efficiency and scalability in environmental purification technologies.

The physical configuration and coordinate system of the unsteady magnetohydrodynamic Sutterby penta-hybrid nanofluid flow over an extending surface are illustrated in Figure 1.

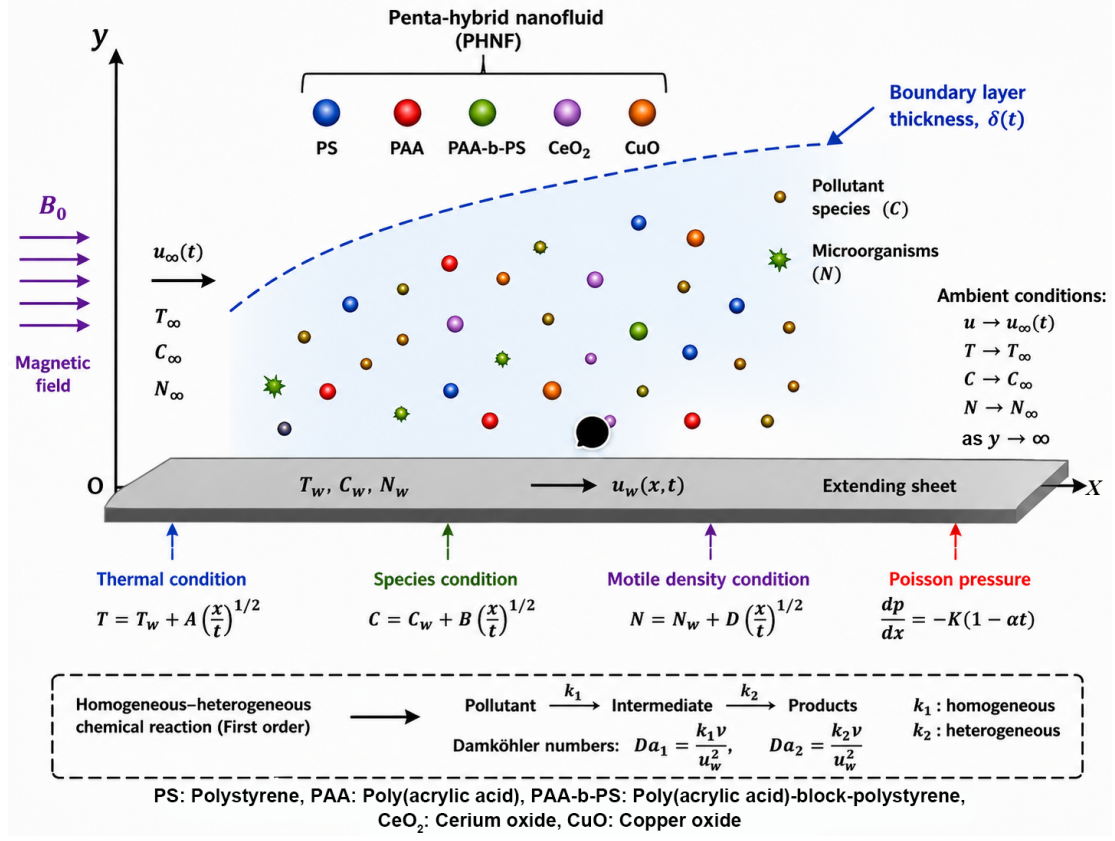


Figure 1. Physical configuration of the unsteady magnetohydrodynamic Sutterby penta-hybrid nanofluid flow over an extending catalytic surface

2 Mathematical Formulation

In the current study, the unsteady magnetohydrodynamic flow of Sutterby penta-hybrid nanofluid over the extended surface for wastewater treatment applications is studied. The mathematical formulation is developed based on established Sutterby-fluid, magnetohydrodynamic nanofluid, and non-Fourier heat and mass transport models reported in the literature [17, 25–27]. The governing equations encompass continuity, momentum, energy, concentration, and Poisson pressure distribution, incorporating shear-dependent viscosity, magnetic forces, and pressure-driven phenomena. Heat transmission is augmented by Brownian motion, thermophoresis, and viscous dissipation, whereas concentration transport includes chemical reactions through Damköhler numbers to account for pollutant degradation.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu_{\text{PHNF}} \left[\frac{\partial^2 u}{\partial y^2} + \frac{mE^2}{2} \left(\frac{\partial u}{\partial y} \right)^2 \frac{\partial^2 u}{\partial y^2} \right] - \frac{\sigma_{\text{PHNF}} B_0^2}{\rho_{\text{PHNF}}} u - \frac{1}{\rho_{\text{PHNF}}} \frac{\partial p}{\partial x} \quad (2)$$

$$\begin{aligned} u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + \lambda_T \left(u^2 \frac{\partial^2 T}{\partial x^2} + v^2 \frac{\partial^2 T}{\partial y^2} + u \frac{\partial u}{\partial x} \frac{\partial T}{\partial x} + v \frac{\partial u}{\partial y} \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \frac{\partial v}{\partial y} + 2uv \frac{\partial^2 T}{\partial x \partial y} + u \frac{\partial T}{\partial y} \frac{\partial v}{\partial x} \right) \\ = \frac{k_{\text{PHNF}}}{(\rho C_p)_{\text{PHNF}}} \frac{\partial^2 T}{\partial y^2} + \frac{\mu_{\text{PHNF}}}{(\rho C_p)_{\text{PHNF}}} \left(\frac{\partial u}{\partial y} \right)^2 + \frac{k_{\text{PHNF}}}{(\rho C_p)_{\text{PHNF}}} \frac{\partial u}{\partial y} + \tau \left[D_B \frac{\partial T}{\partial y} \frac{\partial C}{\partial y} + \frac{D_T}{T_\infty} \left(\frac{\partial T}{\partial y} \right)^2 \right] \\ + \frac{Q_0}{\rho C_p} (T - T_\infty) \end{aligned} \quad (3)$$

$$u \frac{\partial C}{\partial x} + \nu \frac{\partial C}{\partial y} = D_B \frac{\partial^2 C}{\partial y^2} + \frac{D_T}{T_\infty} \frac{\partial^2 T}{\partial y^2} + \lambda_C \left(u^2 \frac{\partial^2 C}{\partial x^2} + \nu^2 \frac{\partial^2 C}{\partial y^2} + u \frac{\partial u}{\partial x} \frac{\partial C}{\partial x} + \nu \frac{\partial u}{\partial y} \frac{\partial C}{\partial x} \right. \\ \left. + 2u\nu \frac{\partial^2 C}{\partial x \partial y} + u \frac{\partial C}{\partial y} \frac{\partial \nu}{\partial x} + \nu \frac{\partial C}{\partial y} \frac{\partial \nu}{\partial y} \right) - k_s C \quad (4)$$

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = -\rho_{\text{PHNF}} \left[\frac{\partial}{\partial x} \left(u \frac{\partial u}{\partial x} + \nu \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial y} \left(u \frac{\partial u}{\partial x} + \nu \frac{\partial u}{\partial y} \right) \right] - \sigma_{\text{PHNF}} B_0^2 \frac{\partial u}{\partial x} \quad (5)$$

Associated boundary conditions are as follows:

$$u = u_w = ax, \quad \nu = \nu_w, \quad T = T_w, \quad \frac{\partial C}{\partial y} = -k_s C_w, \quad P = P_w \quad \text{at } y = 0, \\ u \rightarrow 0, \quad \nu \rightarrow 0, \quad C \rightarrow C_\infty, \quad T \rightarrow T_\infty \quad \text{at } y \rightarrow \infty \quad (6)$$

where, x and y denote the Cartesian coordinates (m) along and normal to the stretching surface, respectively; u and ν are the velocity components ($\text{m}\cdot\text{s}^{-1}$); T and C represent the fluid temperature (K) and pollutant concentration ($\text{kg}\cdot\text{m}^{-3}$), respectively; p is the fluid pressure (Pa); E denotes the electric field strength; σ_{PHNF} is the electrical conductivity ($\text{S}\cdot\text{m}^{-1}$) of the penta-hybrid nanofluid (PHNF); B_0 is the applied magnetic-field strength (T); λ_T and λ_C are the thermal and solutal relaxation parameters, respectively; Q_0 is the volumetric heat-generation coefficient ($\text{W}\cdot\text{m}^{-3}$); k_s is the surface reaction-rate constant (s^{-1}); a is the stretching-rate constant (s^{-1}); u_w and ν_w are the wall velocity components; T_w and C_w denote the wall temperature and concentration; T_∞ and C_∞ represent the ambient values; and P_w denotes the wall-pressure parameter.

The thermophysical properties of the base fluid and the five constituent materials used in the proposed penta-hybrid nanofluid are summarized in Table 1.

Table 1. Physical characteristics of nanoparticles [28–31]

Thermal Properties	ρ ($\text{kg}\cdot\text{m}^{-3}$)	C_p ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$)	κ ($\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$)	σ ($\Omega^{-1}\cdot\text{cm}^{-1}$)
Copper oxide	6500	530–540	18–33	$\sim 10^{-14}$
Cerium oxide	7100–7280	450–550	10–14.4	$\sim 10^{-16}$
Polystyrene	1040–1065	1200–2100	0.10–0.18	$\sim 10^{-16}$
Poly (acrylic acid)	1100–1200	1400–1600	0.32–0.37	$\sim 10^{-16}$
PAA-b-PS	1040–1100	1400–1900	0.12–0.18	$\sim 10^{-16}$

Note: PAA-b-PS = poly (acrylic acid)-block-polystyrene; ρ = corresponding density; C_p = specific heat capacity; κ = thermal conductivity; σ = electrical conductivity.

PAA-b-PS is included as one of the polymeric constituents of the present penta-hybrid nanofluid. It is an amphiphilic diblock copolymer consisting of hydrophilic PAA and hydrophobic PS blocks, which can form a core–corona structure in an aqueous environment [32, 33]. Chockalingam and Natarajan [32] investigated the micellization behaviour of PAA-b-PS in aqueous solution and demonstrated the influence of the block composition on the formation and characteristics of micellar structures. The hydrophobic PS block preferentially associates to form the inner domain, whereas the hydrophilic PAA block interacts with the surrounding aqueous phase and forms the corona, thereby contributing to colloidal stabilization [32, 33]. Since direct thermophysical property data for the specific PAA-b-PS composition considered in this study are not independently available, the effective properties listed in Table 1 were estimated from the corresponding thermophysical properties of its constituent PAA and PS blocks. The effective thermophysical properties were estimated using a composition-weighted approximation based on the corresponding properties of PAA and PS.

The selection of the PAA and PS weight fractions was based on the reported compositional range of PAA-b-PS systems. Fetin et al. [34] reported that a hydrophilic PAA fraction of approximately 25–45 wt% (weight percent) is associated with the formation of vesicular structures in PAA-b-PS systems and showed that variation in the relative block lengths influences the self-assembled morphology and size of the resulting polymeric structures. Accordingly, a PAA weight fraction of $w_{\text{PAA}} = 0.35$ (35 wt%) was selected in the present numerical model, as it lies approximately at the middle of the reported 25–45 wt% hydrophilic range. The corresponding PS weight fraction was therefore taken as $w_{\text{PS}} = 0.65$ (65 wt%). It should be emphasized that the adopted values of 35 wt% PAA and 65 wt% PS are not claimed to represent a universal or experimentally determined composition for PAA-b-PS. Rather, they constitute a representative modelling composition selected within the literature-supported hydrophilic range [34].

Accordingly, the effective value of a thermophysical property X of PAA-b-PS is estimated using the composition-weighted relation $X_{\text{PAA-b-PS}} = w_{\text{PAA}} X_{\text{PAA}} + w_{\text{PS}} X_{\text{PS}}$, where, $X_{\text{PAA-b-PS}}$ denotes the effective thermophysical

property of PAA-b-PS, X_{PAA} and X_{PS} represent the corresponding properties of PAA and PS, respectively, and w_{PAA} and w_{PS} denote their respective weight fractions, with $w_{\text{PAA}} + w_{\text{PS}} = 1$. For the adopted composition, the relation becomes $X_{\text{PAA-b-PS}} = 0.35 X_{\text{PAA}} + 0.65 X_{\text{PS}}$.

The resulting thermophysical properties are therefore treated as estimated effective properties corresponding to the adopted 35:65 PAA/PS composition and are used as modelling parameters in the present penta-hybrid nanofluid formulation.

$$\text{Dynamic viscosity} = D_1 = \frac{\mu_{\text{PHNF}}}{\mu_f} = \frac{1}{(1 - \varphi_1)^{2.5} (1 - \varphi_2)^{2.5} (1 - \varphi_3)^{2.5} (1 - \varphi_4)^{2.5} (1 - \varphi_5)^{2.5}} \quad (7)$$

$$\begin{aligned} \text{Density} = D_2 = \frac{\rho_{\text{PHNF}}}{\rho_f} = (1 - \varphi_5) & \left[(1 - \varphi_4) \left[(1 - \varphi_3) \left[(1 - \varphi_2) \left[1 - \varphi_1 + \varphi_1 \frac{\rho_{s1}}{\rho_f} \right] \right. \right. \right. \\ & \left. \left. + \varphi_2 \frac{\rho_{s2}}{\rho_f} \right] + \varphi_3 \frac{\rho_{s3}}{\rho_f} \right] + \varphi_4 \frac{\rho_{s4}}{\rho_f} \right] + \varphi_5 \frac{\rho_{s5}}{\rho_f} \end{aligned} \quad (8)$$

$$\begin{aligned} \text{Heat Capacity} = D_3 = \frac{(\rho C_p)_{\text{PHNF}}}{(\rho C_p)_f} = (1 - \varphi_5) & \left[(1 - \varphi_4) \left[(1 - \varphi_3) \left[(1 - \varphi_2) \left[1 - \varphi_1 + \varphi_1 \frac{(\rho C_p)_{s1}}{(\rho C_p)_f} \right] \right. \right. \right. \\ & \left. \left. + \varphi_2 \frac{(\rho C_p)_{s2}}{(\rho C_p)_f} \right] + \varphi_3 \frac{(\rho C_p)_{s3}}{(\rho C_p)_f} \right] + \varphi_4 \frac{(\rho C_p)_{s4}}{(\rho C_p)_f} \right] + \varphi_5 \frac{(\rho C_p)_{s5}}{(\rho C_p)_f} \end{aligned} \quad (9)$$

$$\text{Thermal conductivity} = D_4 = \frac{k_{\text{PHNF}}}{k_f} = \frac{k_{s5} + 2A_5 + 2\varphi_5 (k_{s5} - A_5)}{k_{s5} + 2A_5 - \varphi_5 (k_{s5} - A_5)} \quad (10)$$

$$\text{Electrical conductivity} = D_5 = \frac{\sigma_{\text{PHNF}}}{\sigma_f} = \frac{\sigma_{s5} + 2B_5 + 2\varphi_5 (\sigma_{s5} - B_5)}{\sigma_{s5} + 2B_5 - \varphi_5 (\sigma_{s5} - B_5)} \quad (11)$$

where,

$$\begin{aligned} A_2 &= \frac{k_{s1} + 2k_f + 2\varphi_1 (k_{s1} - k_f)}{k_{s1} + 2k_f - \varphi_1 (k_{s1} - k_f)}, & A_3 &= \frac{k_{s2} + 2A_2 + 2\varphi_2 (k_{s2} - A_2)}{k_{s2} + 2A_2 - \varphi_2 (k_{s2} - A_2)}, \\ A_4 &= \frac{k_{s3} + 2A_3 + 2\varphi_3 (k_{s3} - A_3)}{k_{s3} + 2A_3 - \varphi_3 (k_{s3} - A_3)}, & A_5 &= \frac{k_{s4} + 2A_4 + 2\varphi_4 (k_{s4} - A_4)}{k_{s4} + 2A_4 - \varphi_4 (k_{s4} - A_4)}, \\ B_2 &= \frac{\sigma_{s1} + 2\sigma_f + 2\varphi_1 (\sigma_{s1} - \sigma_f)}{\sigma_{s1} + 2\sigma_f - \varphi_1 (\sigma_{s1} - \sigma_f)}, & B_3 &= \frac{\sigma_{s2} + 2B_2 + 2\varphi_2 (\sigma_{s2} - B_2)}{\sigma_{s2} + 2B_2 - \varphi_2 (\sigma_{s2} - B_2)}, \\ B_4 &= \frac{\sigma_{s3} + 2B_3 + 2\varphi_3 (\sigma_{s3} - B_3)}{\sigma_{s3} + 2B_3 - \varphi_3 (\sigma_{s3} - B_3)}, & B_5 &= \frac{\sigma_{s4} + 2B_4 + 2\varphi_4 (\sigma_{s4} - B_4)}{\sigma_{s4} + 2B_4 - \varphi_4 (\sigma_{s4} - B_4)}. \end{aligned}$$

The coefficients $D_1 - D_5$, $A_2 - A_5$, and $B_2 - B_5$ represent the effective thermophysical properties are evaluated using established mixture models. The Brinkman model [35] is used for viscosity, the Maxwell model [36] for thermal conductivity, and conventional mixture relations [37] for density and specific heat. The subscript f denotes the base fluid, whereas $s1, s2, s3, s4$, and $s5$ denote copper oxide, cerium oxide, polystyrene, poly (acrylic acid), and PAA-b-PS, respectively. Their corresponding density (ρ), specific heat capacity (C_p), thermal conductivity (k), dynamic viscosity (μ), and electrical conductivity (σ) are incorporated into the effective-property relations. The nanoparticle volume fractions are defined as φ_1 (copper oxide), φ_2 (cerium oxide), φ_3 (polystyrene), φ_4 (poly (acrylic acid)), and φ_5 (PAA-b-PS).

Introduced similarity variables for a stretching surface are as follows:

$$\begin{aligned} \eta &= \sqrt{\frac{a}{\nu_f}} y, \quad u = ax f'(\eta), \quad \nu = -\sqrt{a\nu_f} f(\eta), \\ \theta &= \frac{T - T_\infty}{T_w - T_\infty}, \quad \phi = \frac{C - C_\infty}{C_w - C_\infty}, \quad p(x, y) = p_\infty + \rho_{\text{PHNF}} a^2 x^2 P(\eta) \end{aligned} \quad (12)$$

where, η is the similarity variable, $f(\eta)$ is the dimensionless stream function, $\theta(\eta)$ is the dimensionless temperature, $\phi(\eta)$ is the dimensionless concentration, and $P(\eta)$ is the dimensionless pressure. The parameter a denotes the stretching rate, ν_f is the kinematic viscosity of the base fluid, and p_∞ is the ambient pressure. The wall and ambient conditions are represented by T_w , T_∞ , C_w , and C_∞ , respectively.

Following the defined transformation, the governing Eqs. (2)–(5) implement non-dimensional form. Using the dimensionless parameters defined in Table 2, the transformed governing equations become:

Table 2. Mathematical form of physical quantities

Dimensionless Parameter	Symbol	Mathematical Definition
Reynolds number	Re	$\frac{ax^2}{\nu_f}$
Deborah number	De	$\frac{E^2 a^2}{\sigma_f B_0^2}$
Magnetic field parameter	B	$\frac{\rho_f a}{\nu_f}$
Prandtl number	Pr	$\frac{a_f}{a^2 x^2}$
Eckert number	Ec	$\frac{C_p(T_f - T_\infty)}{ax\sqrt{\frac{\nu_f}{a}}}$
Peclet number	Pe	$\frac{\lambda_T a}{\tau \frac{D_B(C_w - C_\infty)}{T_\infty \nu_f}}$
Thermal relaxation parameter	Γ_1	$\frac{Q_0}{\rho C_p}$
Brownian motion parameter	N_b	$\frac{D_B}{\nu_f}$
Thermophoresis parameter	N_t	$\frac{\lambda_C a}{k_s}$
Heat generation parameter	Hg	$\frac{k_s}{\sqrt{a/\nu_f}}$
Lewis number	Le	
Solutal relaxation parameter	Γ_2	
Damköhler number	Da	

$$\frac{D_1}{D_2} f''' + \frac{D_1}{D_2} \frac{m}{2} De Re f''^2 f''' + \frac{D_5}{D_2} B^2 f' + f f'' - f'^2 - 2P' = 0 \quad (13)$$

$$\frac{D_4}{D_3} \frac{1}{Pr} \theta'' + f \theta' - \Gamma_1 (f^2 \theta'' + f f' \theta') + \frac{D_1}{D_3} Ec (f'')^2 + \frac{D_4}{D_3} \frac{Pe}{Pr} + (N_b \theta' \phi' + N_t \theta'^2) + Hg \theta = 0 \quad (14)$$

$$\phi'' + Le f \phi' + \frac{N_t}{N_b} \theta'' - \Gamma_2 (f^2 \phi'' + f f' \phi') = 0 \quad (15)$$

$$P'' + \frac{1}{Re} (2f'^2 + B^2 f') = 0 \quad (16)$$

Associated boundary conditions are as follows:

$$f(0) = f_w, f'(0) = 1, \theta(0) = 1, \phi'(0) = -Da, P(0) = P_w, f(\infty) = 0, \theta(\infty) = 0, \phi(\infty) = 0, P(\infty) = 0 \quad (17)$$

3 Method of Solution

The outcomes of the mathematical model outlined in Eqs. (13)–(16), along with the boundary conditions in Eq. (17), are determined using the bvp5c method in MATLAB. The procedure detailed below is employed to obtain the solutions:

$$\begin{aligned} f &= h_1, & f' &= h_2, & f'' &= h_3, & f''' &= h'_3, & \theta &= h_4, & \theta' &= h_5, & \theta'' &= h'_5, \\ \phi &= h_6, & \phi' &= h_7, & \phi'' &= h'_7, & P &= h_8, & P' &= h_9, & P'' &= h'_9 \end{aligned} \quad (18)$$

$$h'_1 = h_2 \quad (19)$$

$$h'_2 = h_3 \quad (20)$$

$$h'_3 = \frac{-\frac{D_2}{D_1} \left[\frac{D_5}{D_2} B^2 h_2 + h_1 h_3 - h_2^2 - 2h_9 \right]}{1 + \frac{m}{2} De Re h_3^2} \quad (21)$$

$$h'_4 = h_5 \quad (22)$$

$$h'_5 = - \frac{\left[h_1 h_5 - \Gamma_1 (h_1 h_2 h_5) + \frac{D_1}{D_3} E c h_3^2 + \frac{D_4 P_e}{D_3 P_r} + (N_b h_5 h_7 + N_t h_5) + H g h_4 \right]}{\frac{D_4}{D_3 P_r} - \Gamma_1 h_1^2} \quad (23)$$

$$h'_6 = h_7 \quad (24)$$

$$h'_7 = - \frac{\left[L e h_1 h_5 + \frac{N_t}{N_b} h'_5 - \Gamma_2 (h_1 h_2 h_7) \right]}{1 - \Gamma_2 h_1^2} \quad (25)$$

$$h'_8 = h_9 \quad (26)$$

$$h'_9 = - \frac{1}{Re} (2h_2^2 + B^2 h_2) \quad (27)$$

Boundary conditions become the following:

$$\begin{aligned} h_1(0) = f_w, \quad h_2(0) = 1, \quad h_4(0) = 1, \quad h_7(0) = -Da h_6(0), \quad h_8(0) = P_w, \\ h_1(\infty) = 0, \quad h_4(\infty) = 0, \quad h_6(\infty) = 0, \quad h_8(\infty) = 0 \end{aligned} \quad (28)$$

4 Result and Discussion

Figures 2–5 show the effects of key parameters on the velocity, temperature, concentration, and pressure profiles, respectively.

4.1 Velocity Profile

Figure 2a shows the influence of the magnetic field parameter (B) on the dimensionless velocity profile for a hybrid nanofluid consisting of polystyrene, poly (acrylic acid), PAA-b-PS, cerium oxide, and copper oxide. The figure shows that an increase in B from 0.5 to 0.8 results in the reduction in the velocity of the fluid throughout the boundary layer of the surface. This occurs because the applied transverse magnetic field induces a Lorentz force through its interaction with the electrically conducting penta-hybrid nanofluid, generating an electromagnetic drag that acts as a resistive body force and opposes fluid motion. A stronger magnetic field produces greater magnetohydrodynamic drag. This results in thickening the velocity boundary layer and decelerating the flow. The deceleration by the parameter B extends the residence time of the contaminated fluid within the active catalytic zone, in which the copper oxide and cerium oxide nanoparticle system allows more time to achieve photocatalytic oxidation, adsorption, and microbial inactivation. The polymers such as polystyrene, poly (acrylic acid), and PAA-b-PS further amplify this retardation near the stretching surface through viscoelastic resistance and chain entanglement effect which affect the total flow resistance. Additionally, the incremental addition of the nanoparticle volume fractions reduces the velocity profile for both B values due to the enhanced effective viscosity of the nanofluids. These results confirm that the parameter B , nanoparticle loading and polymer loading suppress the momentum transport in the boundary layer.

From a practical wastewater treatment standpoint, the applied magnetic field can then be used as an external flow control mechanism. Application of a modest magnetic field can slow the transit velocity and increase the residence duration of the polluted fluid in the catalytic region, which results in the enhancement of the pollutant-catalyst contact. However, excessive magnetic retardation may lower the total treatment throughput. Therefore, the magnetic field should be optimized to balance residence duration and flow rate in continuous-flow treatment systems.

The effect of the Deborah number (De) on the velocity profile is illustrated in Figure 2b. It is observed that increasing De from 0.3 to 0.6 increases the dimensionless velocity profile throughout the boundary layer. Under the present parameter configuration, a higher De represents a stronger viscoelastic response associated with an increased ratio of fluid relaxation time to the characteristic flow time. The additional elastic stresses modify the streamwise momentum transfer, while the interaction of cerium oxide and copper oxide nanoparticles with the polymer chains

influences the overall rheological response of the penta-hybrid nanofluid. The polystyrene, poly (acrylic acid), and PAA-b-PS chains introduce molecular memory and chain-interaction effects, which modify the momentum transport characteristics. Under the present numerical configuration, these coupled viscoelastic and nanoparticle polymer interactions result in an enhancement of the computed velocity profile as De increases. This trend is consistent with Figure 2b, where the $De = 0.6$ curve lies above the $De = 0.3$ curve throughout the boundary-layer region.

Figure 2c depicts the influence of the Reynolds number (Re) on the dimensionless velocity profile $f'(\eta)$. It is observed that increasing Re decreases the velocity profile throughout the momentum boundary layer. Physically, a larger Re intensifies inertial effects relative to viscous diffusion, thereby reducing the momentum boundary-layer thickness and suppressing the dimensionless velocity profile. Figure 2d illustrates how the wall suction parameter (f_w) modifies the velocity profile, evaluated at $f_w = 0.5$ and $f_w = 0.8$. As f_w increases, the velocity profile decreases and the boundary layer region becomes thinner. The added contribution of cerium oxide and copper oxide nanoparticles embedded within the polystyrene, poly (acrylic acid), and PAA-b-PS matrices elevates the effective viscosity and density, further reducing the velocity magnitude. Higher nanoparticle loading amplifies this viscous resistance effect. Together, the suction intensity and nanoparticle concentration serve as independent yet complementary regulators of the velocity distribution and boundary layer stability within the penta-hybrid nanofluid system.

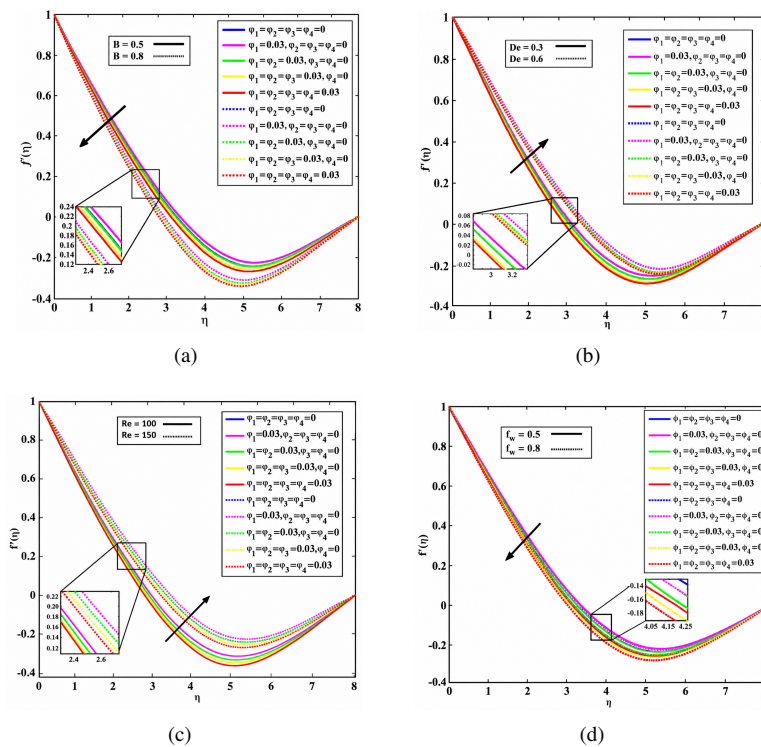


Figure 2. Effects of (a) magnetic field parameter (B); (b) Deborah number (De); (c) Reynolds number (Re); and (d) wall suction parameter (f_w) on the velocity profile

4.2 Temperature Profile

The effect of the Eckert number (Ec) on the temperature distribution is shown in Figure 3a. The mechanical energy is irreversibly dissipated into thermal energy through viscous friction, a process amplified by the high effective viscosity transmitted by the poly (acrylic acid) and polystyrene polymer chains, which intensify internal frictional heating because the increase in Ec elevates the temperature throughout the boundary layer. The high thermal conductivity facilitates heat diffusion within this thermophoretic transport process by the nanoparticles. The resulting temperature elevation is beneficial in the wastewater treatment, as it increases the photocatalytic and redox reactivity of cerium oxide and copper oxide nanoparticles, thereby accelerating the degradation kinetics of organic pollutants. The polymers increase the viscoelasticity of the fluid, reduce nanoparticle agglomeration, improve nanoparticle stability, and promote more uniform thermal transport, thereby providing a more favorable thermal environment for catalytic reactions in wastewater treatment.

Figure 3b illustrates the influence of the thermal relaxation parameter (Γ_1) on the temperature profile $\theta(\eta)$. It is observed that increasing Γ_1 reduces the temperature distribution throughout the thermal boundary layer. Physically,

Γ_1 accounts for the finite time required for heat to propagate within the fluid according to the Cattaneo–Christov heat-flux model. As Γ_1 increases, the delay in heat transport becomes more pronounced, thereby weakening thermal diffusion and reducing the thermal boundary-layer thickness. Consequently, the fluid temperature decreases with increasing Γ_1 .

The effect of the Peclet number (Pe) on the temperature distribution is illustrated in Figure 3c. As Pe increases from 0.2 to 0.5, the temperature profile decreases within the thermal boundary layer because convective heat transport becomes dominant over thermal diffusion, which removes the heat from the wall region fast, creating a thinner thermal boundary layer. The effective thermal conductivity and thermal diffusivity of fluids in the hybrid nanofluid system strengthen convective heat dissipation and reduce the temperature distribution as nanoparticle loading rises because of the nanoparticles. As a result, the nanofluid that is fully loaded shows the most effective heat removal behavior. Meanwhile, the catalytic performance and effective heat management are ensured by the PAA-b-PS stabilizer by maintaining nanoparticle dispersion and preventing thermal degradation or agglomeration during high-temperature operation. It shows that reducing excess exothermic reaction heat and preserving thermal stability in catalytic wastewater treatment systems can be attained by regulating Pe through flow velocity and nanofluid composition.

Hence, in practice, Pe can be adjusted by the flow velocity and characteristic parameters of the treatment channel in reactor design. A higher Pe is in support of rapid convective heat removal and may help to prevent excessive temperature build-up. A smaller value allows for stronger thermal diffusion in the catalytic zone. Thus, a suitable balance between convection and diffusion is necessary for maintaining favorable temperature conditions for the catalytic breakdown of pollutants.

Figure 3d shows the influence of the heat generation parameter (Hg) on the temperature profile. Since Hg is a measure of the strength of internal volumetric heat generation in the hybrid nanofluid system, increasing it from 3 to 5 considerably raises the temperature profile and thickens the thermal boundary layer. The increased temperature rise at the surface is caused by heat-producing processes that connect cerium oxide and copper oxide nanoparticles which include photocatalytic energy thermalization, exothermic adsorption and redox reactions, electromagnetic induction heating, and pollutant–surface electron transfer processes. Higher nanoparticle loading increases the number of active heat-generating sites, which amplifies this impact even more. Furthermore, the PAA-b-PS matrix provides additional thermal energy through viscoelastic dissipation while stabilizing the nanoparticles against sintering and agglomeration. Finally, the fully loaded hybrid nanofluid at $Hg = 5$ has the deepest thermal penetration and the highest temperature distribution, which enhances the catalytic pollutant degradation through Arrhenius thermal activation. Even if heat generation is too high, it may cause polymer degradation and catalyst deactivation at last.

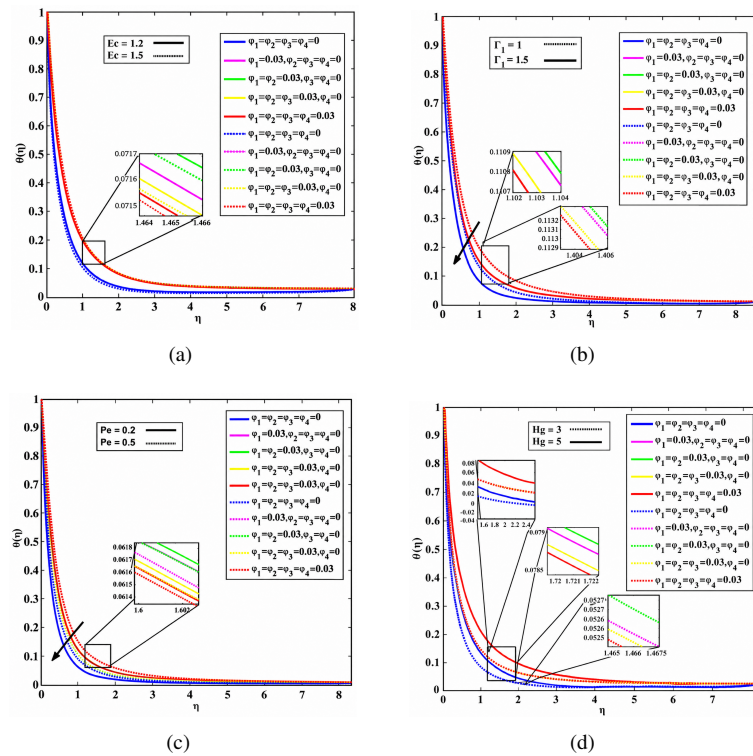


Figure 3. Effects of (a) Eckert number (Ec); (b) thermal relaxation parameter (Γ_1); (c) Peclet number (Pe); and (d) heat generation parameter (Hg) on the temperature profile

4.3 Concentration Profile

Figure 4a presents the effects of the Lewis number (Le) on the concentration profile for poly (acrylic acid)/ polystyrene/PAA-b-PS/cerium oxide/copper oxide hybrid nanofluid system. The solutal boundary layer thickness is reduced by increasing Le which lowers the mass diffusivity relative to the momentum diffusivity, concentrating the species gradient closer to the catalytic wall. Within this thinner solutal layer, the polymer matrices, particularly polystyrene and poly (acrylic acid), facilitate pollutant adsorption by maintaining close proximity between contaminant molecules and the reactive cerium oxide and copper oxide nanoparticles. Increasing Damköhler number (Da) intensifies catalytic reaction rates well beyond the rate of diffusive mass resupply, resulting in a sharp reduction in local concentration throughout the boundary layer. The photocatalytic activity of cerium oxide and the oxidative and bactericidal properties of copper oxide are the primary kinetic drivers of this reaction-dominated behavior. Together, reduced mass diffusion (high Le) and intensified surface reactions (high Da) produce substantially more complete contaminant removal, validating the dual-mechanism design of the penta-hybrid system.

Figure 4b examines the effect of Da on the concentration profile. The concentration decreases consistently across the boundary layer when Da increases, confirming that chemical reaction kinetics control over diffusive mass transport in this regime validates the incorporation of catalytic surface reaction terms in the model. This behavior provides direct support for the design of catalytic filter beds incorporating cerium oxide and copper oxide nanoparticles, in which low permeability can prolong the contact between pollutants and the catalytic surface. The functional groups present on poly (acrylic acid) (carboxyl) and polystyrene (aromatic) facilitate adsorption and anchoring of target pollutants, ensuring sustained interfacial contact essential for effective degradation of the pollutants. The synergy between Da -driven reaction enhancement confirms the integrated design advantage of combining chemically reactive nanoparticles with polymer stabilizers in a penta-hybrid configuration.

From an operational point of view, these results imply that the removal of pollutants is a function of the proper balance between mass transfer and reaction kinetics. Le is a measure of the relative diffusion of the contaminating species, and Da is a measure of the relative strength of the chemical reaction. A larger Da results in a faster consumption of contaminants. However, in order to have an efficient treatment, there should be enough mass transit to constantly provide pollutants to the active catalytic surface. Therefore, reactor residence time, flow rate, catalyst loading, and mass transfer conditions should be optimized simultaneously, rather than focusing solely on increasing the reaction rate.

Figure 4c illustrates the effect of the solutal relaxation parameter (Γ_2) (ranging from 0.1 to 0.3) on the dimensionless concentration profile $\phi(\eta)$. The concentration profile is influenced by the increase in Γ_2 , which represents the finite relaxation time associated with solutal transport. As Γ_2 increases, the response of the concentration field to changes in the surrounding flow becomes delayed, thereby modifying the mass-transfer process and concentration boundary-layer thickness. Thus, the observed variation in $\phi(\eta)$ can be attributed to non-Fickian solutal relaxation and the associated mass-transport delay. The catalytic reaction rate is governed separately by Da and the corresponding reaction term in the governing concentration equation.

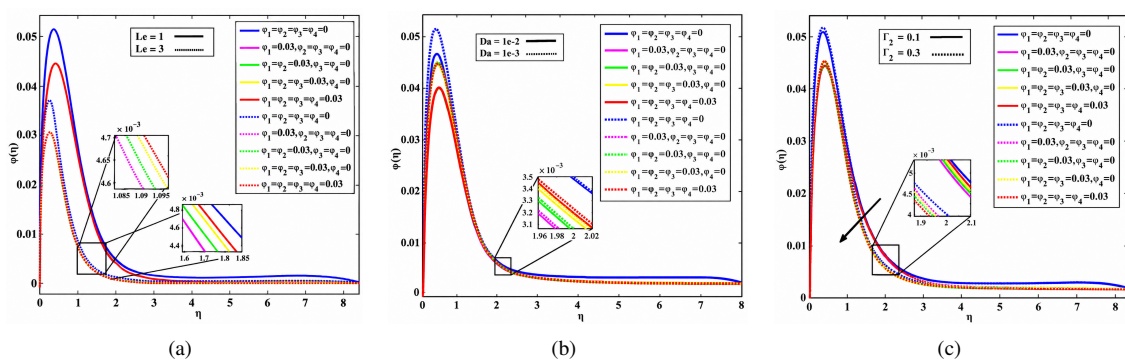


Figure 4. Effects of (a) Lewis number (Le); (b) Damköhler number (Da); and (c) solutal relaxation parameter (Γ_2) on the concentration profile

4.4 Pressure Profile

Figure 5a illustrates the effect of the magnetic field parameter. The application of a magnetic field generates a Lorentz force that opposes fluid motion, thereby increasing the resistance to flow and modifying the pressure distribution. As the parameter B increases from 0.3 to 0.5, the pressure within the boundary layer increases. The effective electrical conductivity of fluids in the hybrid nanofluid is improved by the nanoparticles, which

strengthens the magnetic connection and improves pressure building as nanoparticle loading rises. The equilibrium between inertial flow effects and magnetic retardation results in a bell-shaped pressure profile, with effective peaking pressure in the boundary layer and then decreasing in the far field. Meanwhile, the magnetohydrodynamic behavior and homogeneous electrical conductivity are achieved by the PAA-b-PS stabilizer by preventing agglomeration and maintaining uniform nanoparticle dispersion. The projected variation in pressure suggests that in practical membrane-based wastewater treatment systems, magnetic and flow characteristics may affect the pressure conditions required to convey fluid through the treatment region. Suitable management of the pressure distribution may facilitate pollutant–catalyst interaction and membrane permeability. However, too high pressure may raise the operational energy need and the membrane stress; hence, the pressure field should be optimized combined with the flow rate and treatment efficiency.

Figure 5b illustrates the effect of the Re on the pressure profile. As Re increases from 100 to 150, the pressure profile decreases throughout the boundary layer, with the $Re = 150$ curve remaining below the $Re = 100$ curve. This indicates that higher Re produces a lower dimensionless pressure response in the present formulation. The observed reduction is associated with the relative scaling of inertial and viscous contributions in the similarity-transformed momentum equation, which modifies the pressure required to satisfy the momentum balance. Therefore, for the present parameter range, increasing Re from 100 to 150 reduces the pressure distribution, whereas lower Re ($Re = 100$) corresponds to the higher pressure profile.

Figure 5c represents the effect of the wall pressure parameter (P_w). Stronger wall injection includes more fluid mass to the boundary layer that raises internal pressure to maintain flow continuity, resulting in a wall-normal momentum effect. This causes the pressure distribution to rise noticeably when P_w is increased from 0.2 to 0.4. The addition of cerium oxide and copper oxide nanoparticles increases the pressure by raising the effective density and viscosity of the nanofluid. In addition, the polymers such as polystyrene, poly (acrylic acid), and PAA-b-PS components contribute viscoelastic normal stresses that protect from deformation during wall injection and produce additional elastic pressure effects. Therefore, the fully loaded hybrid nanofluid at higher P_w shows the highest pressure profiles. This is useful for membrane wastewater treatment because the higher pressure can simultaneously support catalytic degradation of pollutants during the backwashing process, thereby improving membrane cleaning efficiency and reducing fouling.

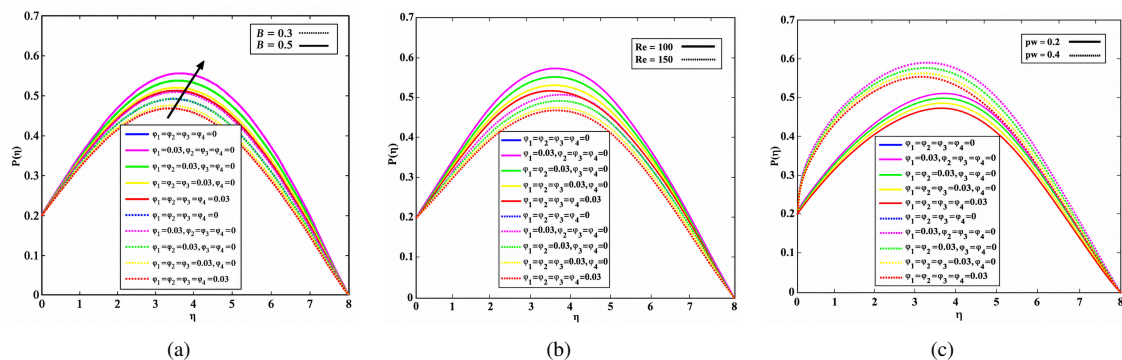


Figure 5. Effects of (a) magnetic field parameter (B); (b) Reynolds number (Re); and (c) wall pressure parameter (P_w) on the pressure profile

5 Conclusion

This study numerically investigated the unsteady magnetohydrodynamic flow, heat transfer, pollutant concentration, and pressure distribution of a Sutterby penta-hybrid nanofluid over an extended surface using the MATLAB bvp5c method. The model incorporates the effects of magnetic fields, viscoelasticity, thermal and solutal relaxation, heat generation, mass diffusion, chemical reactions, and the volume fractions of five penta-hybrid nanofluid constituents. The following conclusions are obtained from the computed profiles:

- **Velocity profile:** Increasing the parameter B from 0.5 to 0.8 reduces the dimensionless velocity because of the additional magnetohydrodynamic resistance. Increasing De from 0.3 to 0.6 increases the velocity profile, consistent with the plotted trend. The Re variation also requires consistent interpretation with the plotted solution; therefore, the reported trend is retained only where it agrees with Figure 2c. Increasing the suction parameter reduces the velocity and compresses the momentum boundary layer.
- **Temperature profile:** Increasing Γ_1 from 1.0 to 1.5 moderately increases the temperature and thickens the thermal boundary layer. Increasing Pe from 0.2 to 0.5 decreases the temperature profile and reduces the

thermal boundary-layer thickness. An increase of Hg from 3 to 5 produces a higher temperature profile and a thicker thermal boundary layer.

- **Concentration profile:** Increasing Le reduces the concentration boundary-layer thickness, indicating a reduction in the relative mass-diffusion effect. Increasing Da decreases the concentration profile, demonstrating the stronger influence of chemical reaction relative to diffusive transport in the considered regime. Γ_2 is interpreted as a measure of non-Fickian mass-transport relaxation and delay rather than as a direct measure of catalytic reaction rate. For the fully loaded penta-hybrid nanofluid, $\varphi_1 = \varphi_2 = \varphi_3 = \varphi_4 = \varphi_5 = 0.03$, and the concentration approaches zero before $\eta = 4$ for the reported $\Gamma_2 = 0.3$ case.
- **Pressure profile:** Increasing the magnetic parameter from 0.3 to 0.5 increases the pressure profile. More importantly, Figure 5b shows that increasing Re from 100 to 150 produces a lower pressure profile, with the $Re = 150$ curve below the $Re = 100$ curve. Thus, the Re effect is reported in the study strictly according to the plotted numerical result. Increasing the Pw from 0.2 to 0.4 increases the pressure distribution.

Overall, the results demonstrate that the governing dimensionless parameters regulate the momentum, thermal, mass-transfer, and pressure characteristics of the Sutterby penta-hybrid nanofluid. In particular, magnetic field strength and suction provide mechanisms for modifying the velocity and boundary-layer development, while Pe , Le , Da , Γ_1 , and Γ_2 influence thermal and solutal transport. The nanoparticle volume fractions also modify the computed transport profiles, with the fully loaded case corresponding to $\varphi_1 = \varphi_2 = \varphi_3 = \varphi_4 = \varphi_5 = 0.03$.

The above findings provide a qualitative numerical basis for considering magnetohydrodynamic penta-hybrid nanofluids in catalytic wastewater-treatment configurations. The results suggest that flow velocity, thermal transport, mass diffusion, chemical reaction, and pressure should be considered simultaneously when selecting operating conditions. However, the present study is numerical, and therefore the implications for membrane permeability, pollutant-removal efficiency, energy consumption, and reactor performance should be regarded as potential engineering applications rather than experimentally established outcomes. Experimental and/or pilot-scale investigations are required to quantify these effects under practical operating conditions.

Author Contributions

Conceptualization, A.P. and V.P.P.; methodology, A.P.; software, A.P.; validation, S.R.P.M., Y.S., and S.M.K.; formal analysis, V.P.P.; investigation, S.M.K.; data curation, S.R.P.M.; writing—original draft preparation, V.P.P.; writing—review and editing, A.P. and Y.S.; visualization, V.P.P.; supervision, A.P.; project administration, A.P. and S.M.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used generative AI in order to improve the language of the article. After using this tool or service, the authors reviewed and edited the content as needed and took full responsibility for the publication.

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